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LINUX_ASSIGNMENT_05

1. Write a script to calculate factorial of a number using while loop.

*shell.sh ×

```
#!/bin/bash
read -p "Enter a number: " n
fact=1
i=1
while (( $i <= $n ))
do
fact=$((fact*i))
i=$((i+1))
done
echo "Factorial = $fact"
```

2. Write a script to display first 20 even numbers using for loop.

*shell.sh ×

```
#!/bin/bash

for ((i=2, i<=40, i+=2)) do
    echo $i
    done
```

3. Write a script to print the multiplication table of a number given by the user.

*shell.sh ×

```
#!/bin/bash

read -p "Enter the number: " num

for ((i=1; i<=10; i++)) do
    value = $((num * i))
    echo $value
    done
```

4. **Write a script to count lines, words, and characters in a file.**

*shell.sh ×

```
#!/bin/bash
read -p "Enter file name: " f
wc $f
```

5. **Write a script to display a menu (date, calendar, list files, exit) and execute the chosen option.**

*shell.sh ×

```
#!/bin/bash
echo "1. Date"
echo "2. Calendar"
echo "3. List Files"
echo "4. Exit"
read -p "Choose option: " ch

case $ch in
1) date ;;
2) cal ;;
3) ls ;;
4) exit ;;
*) echo "Invalid choice" ;;
esac
```

(Linux Shell Programming continued)

1. Write a shell script that accepts a filename as an argument and displays whether it is a directory, a regular file, or something else.

*shell.sh ×

```
#!/bin/bash

if [ -d "$1" ]
then
echo "It is a directory"
elif [ -f "$1" ]
then
echo "It is a regular file"
else
echo "It is something else"
fi
```

2. **Write a script that reads a number and prints whether it is positive, negative, or zero.**

*shell.sh ×

```
#!/bin/bash

read -p "Enter a number: " num

if (( $num > 0 ))
then
echo "Positive"
elif (( $num < 0 ))
then
echo "Negative"
else
echo "Zero"
fi
```

3. **Write a script that accepts a username and checks whether the user is currently logged in.**

*shell.sh ×

```
#!/bin/bash

read -p "Enter username: " user

who | grep "$user"

if (( $? == 0 ))
then
echo "User is logged in"
else
echo "User is not logged in"
fi
```

4. **Write a script that prints the first 10 natural numbers using until loop.**

*shell.sh ×

```
#!/bin/bash

i=1

until (( $i > 10 ))
do
echo $i
((i++))
done
```

5. Write a script that demonstrates use of functions by creating one for addition and one for subtraction.

*shell.sh ×

```
#!/bin/bash

add() {
echo "Addition = $(($1 + $2))"
}

sub() {
echo "Subtraction = $(($1 - $2))"
}

read -p "Enter first number: " a
read -p "Enter second number: " b

add $a $b
sub $a $b
```